Trifunctional Crosslinking Modeling

Your Name

Your Institution

November 21, 2025

Outline

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6. Summary

Experimental Context

- In crosslinking experiments, two types of crosslinks form:
  - **Trifunctional crosslinks**: All three reactive ends connected
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- Both types occur simultaneously in the same experiment
- **Key challenge**: Model both types on equal footing

Modeling Requirement

Need unified framework to account for both bifunctional and trifunctional crosslinks

Total Number of Crosslinks

Crosslink Decomposition

Total number of crosslinks formed in experiment:

$$N_{XL} = N_{XL}^{\text{tri}} + N_{XL}^{\text{bi}} \tag{1}$$

- N_{XL}: Total observed crosslinks
- N_{XL}^{tri}: Number of trifunctional crosslinks
- N_{XL}^{bi}: Number of bifunctional crosslinks

The β Parameter

Definition

Given N_L lysines in the system:

$$\beta = \frac{N_{XL}}{N_{LP} + N_{LT}} = \frac{N_{XL}^{\text{tri}} + N_{XL}^{\text{bi}}}{N_{LP} + N_{LT}} \tag{2}$$

- N_{LP}: Number of lysine pairs
- N_{LT}: Number of lysine triplets

Physical Interpretation

β incorporates:

- Structural constraints
- Chemical reactivity (not all proximal lysines react)

Combinatorial Expressions

Counting Pairs and Triplets

$$N_{LP} = \frac{N_L(N_L - 1)}{2} \tag{3}$$

$$N_{LT} = \frac{N_L(N_L - 1)(N_L - 2)}{6} \tag{4}$$

- N_{LP}: Choose 2 lysines from N_L (order doesn't matter)
- N_{LT}: Choose 3 lysines from N_L (order doesn't matter)

Simplified β Formula

Derivation

Substituting combinatorial expressions:

$$\beta = \frac{N_{XL}^{\text{tri}} + N_{XL}^{\text{bi}}}{\frac{N_L(N_L-1)}{2} + \frac{N_L(N_L-1)(N_L-2)}{6}} \tag{5}$$

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Simplified Form

$$\boxed{\beta = \frac{6\left(N_{XL}^{\text{tri}} + N_{XL}^{\text{bi}}\right)}{N_L(N_L-1)(N_L+1)}} \tag{6}$$

Factor of 6 comes from common denominator manipulation

Definition of α

Observable vs True Crosslinks

$$\alpha = \frac{N_{XL}^{\text{obs,T}}}{N_{XL}^{\text{obs}}} = \frac{N_{XL}^{\text{obs,tri,T}} + N_{XL}^{\text{obs,bi,T}}}{N_{XL}^{\text{obs,tri}} + N_{XL}^{\text{obs,bi}}} \tag{7}$$

- $N_{XL}^{\text{obs,T}}$: True positive crosslinks (actually formed)
- N_{XL}^{obs}: Total observed crosslinks
- α represents the **true positive rate**

Parameter Bounds

Upper Bound for α/β

$$\frac{\alpha}{\beta} \leq \frac{N_L(N_L-1)(N_L+1)}{N_{XL}^{\text{obs,tri}} + N_{XL}^{\text{obs,bi}}} \tag{8}$$

Upper Bound for $(1-\alpha)/(1-\beta)$

$$\frac{1-\alpha}{1-\beta} \leq \frac{N_L(N_L-1)(N_L+1)}{N_{XL}^{\text{obs,tri}} + N_{XL}^{\text{obs,bi}}} \tag{9}$$

These bounds constrain the feasible region for α and β values

System Specification

System Parameters

- $N_L = 100$ (total lysines)
- $N_{XL}^{\text{obs}} = 100$ (observed crosslinks)
- $N_{XL}^{\text{obs,tri}} = 20$ (observed trifunctional)
- $N_{XL}^{\text{obs,bi}} = 80$ (observed bifunctional)

Simplifying Assumptions

1. All crosslinks have the **same forward model probability**
2. All crosslinks share the **same** α **and** β parameters

Total Likelihood

Likelihood for Observation O_n

$$P(O_n|X) = \frac{\alpha}{\beta} P(XL_n|X) + \frac{1-\alpha}{1-\beta} P(\overline{XL_n}|X) \tag{10}$$

- $P(XL_n|X)$: Probability crosslink n is satisfied by structure X
- $P(\overline{XL_n}|X)$: Probability crosslink n is NOT satisfied
- First term weighted by α/β (true positives)
- Second term weighted by $(1-\alpha)/(1-\beta)$ (false positives)

Factorized Forward Model

Decomposition by Crosslink Type

An observed crosslink can be either bifunctional or trifunctional:

$$\boxed{P(XL_n|X) = P(XL_n^{\text{tri}}|XL_n)P(XL_n^{\text{tri}}|X) + P(XL_n^{\text{bi}}|XL_n)P(XL_n^{\text{bi}}|X)} \quad (11)$$

- $P(XL_n^{\text{tri}}|XL_n)$: Probability it's trifunctional given it's a crosslink
- $P(XL_n^{\text{tri}}|X)$: Structural probability for trifunctional
- $P(XL_n^{\text{bi}}|XL_n)$: Probability it's bifunctional given it's a crosslink
- $P(XL_n^{\text{bi}}|X)$: Structural probability for bifunctional

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   - Forward model probabilities $P(XL|X)$
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2. Two key parameters:
   - β: Accounts for structural and chemical reactivity constraints
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3. Likelihood function combines:
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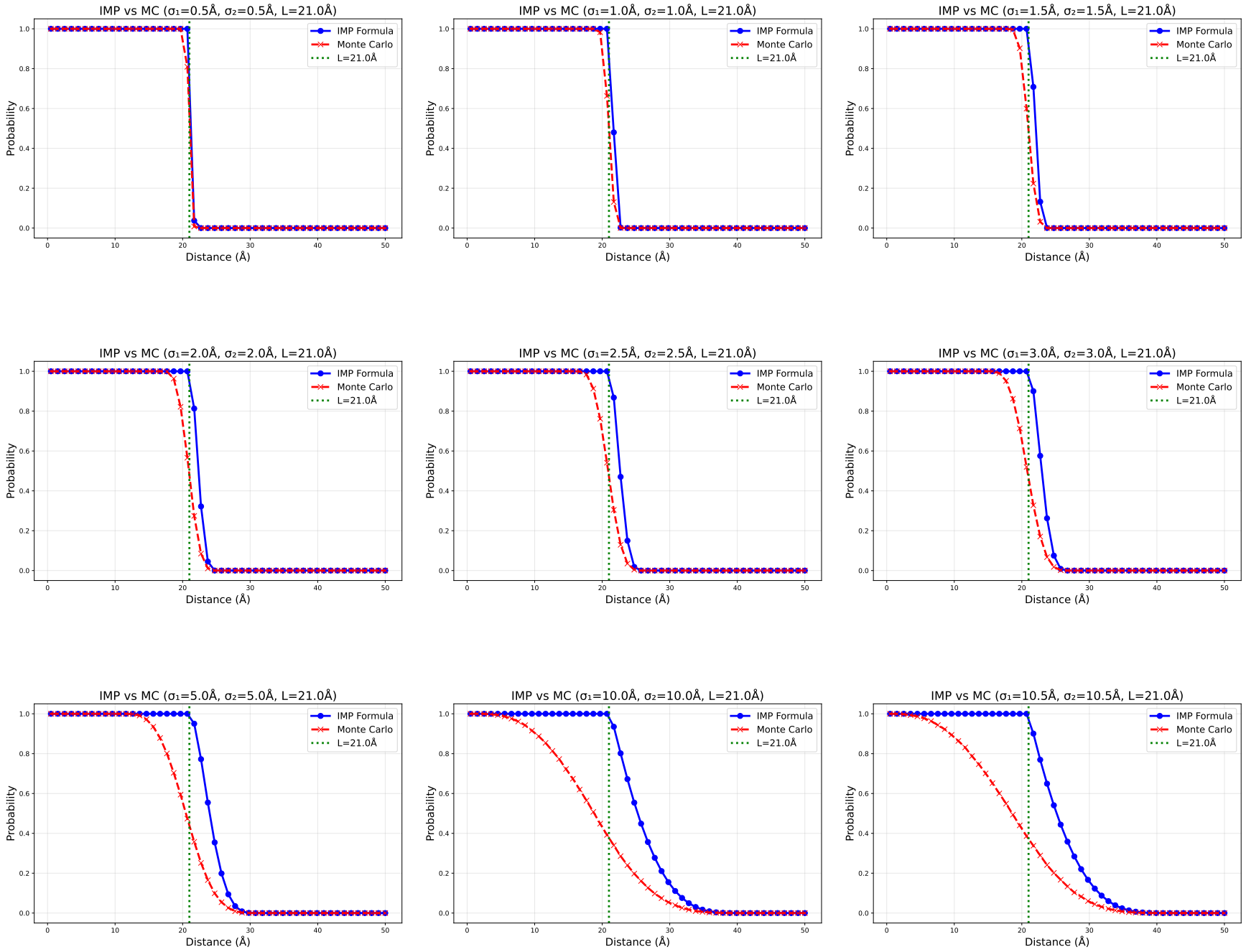
4. Forward model factorizes by crosslink type

Next Steps

Implement Bayesian inference to estimate α, β, and structural ensemble X


IMP vs MC (σ₁=0.5Å, σ₂=0.5Å, L=21.0Å)
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IMP Formula
Monte Carlo
L=21.0Å
Probability
Distance (Å)